Implementation of Regenerative Braking in BLDC Motor with Trapezoidal Control

Dr. Subramaniam K
School of Electrical Engineering
Vellore Institute of Technology,
Vellore

Abinaya S
School of Electrical Engineering,
Vellore Institute of Technology,
Vellore

Hannah Ajish
School of Electrical Engineering,
Vellore Institute of Technology
Vellore

***Abstract*— This paper presents the modelling, design, and simulation of a Vehicle Braking Control System (VBCS) to enhance vehicle stability and reduce stopping distance under critical braking conditions. The primary objective was to develop a controller capable of maintaining optimal wheel slip during emergency stops, thereby maximizing tire-road friction without causing wheel lock-up. The entire system—including the electronic controller, hydraulic actuation, and quarter-car vehicle dynamics model—was implemented and rigorously simulated within the Simulink/MATLAB environment. Simulation results, utilizing a realistic brake pedal input, demonstrate the controller's effectiveness in achieving rapid deceleration and precise slip ratio regulation compared to uncontrolled braking. The findings validate the proposed VBCS architecture as a robust solution for improving vehicle safety and performance, providing a foundational step toward real-world Electronic Braking Systems (EBS) or Anti-lock Braking Systems (ABS) implementation.**

Keywords—Trapezoidal, PID, BLDC, Regenerative Braking

I. INTRODUCTION

This paper presents the modelling, design, and simulation of a Vehicle Braking Control System (VBCS) to enhance vehicle stability and reduce stopping distance under critical braking conditions. The primary objective was to develop a controller capable of maintaining optimal wheel slip during emergency stops, thereby maximizing tire-road friction without causing wheel lock-up. The entire system—including the electronic controller, hydraulic actuation, and quarter-car vehicle dynamics model—was implemented and rigorously simulated within the Simulink/MATLAB environment. Simulation results, utilizing a realistic brake pedal input, demonstrate the controller's effectiveness in achieving rapid deceleration and precise slip ratio regulation compared to uncontrolled braking. The findings validate the proposed VBCS architecture as a robust solution for improving vehicle safety and performance, providing a foundational step toward real-world Electronic Braking Systems (EBS) or Anti-lock Braking Systems (ABS) implementation.

II. LITERATURE REVIEW

A. Commutation and Speed Control Architectures

The intrinsic trapezoidal BEMF profile makes it possible for BLDC motors[1] to adopt the simple trapezoidal commutation (or six-step commutation) method directly, in contrast with the vector-based control in PMSMs. This basic current control strategy has been discussed and implemented in numerous works, focusing on its efficiency and implementation.

Rex et al. (2023)[5] presented a base analysis by explaining closed-loop speed control of the BLDC motor, using the trapezoidal commutation technique in conjunction with a PI controller and PWM. The work of the scientists confirmed that this is one of the sensor-based approaches which can be used for robust regulation of speed and provides the backbone in many industrial and automotive drives. Extending the discussion to the aspect of implementation, a sensor-based trapezoidal control scheme was demonstrated to be practically viable by Jhawar et al. (2024) [9]. Their hardware design was developed on an XMC-7200 microcontroller and particularly aimed at applications involving EVs, confirming the simple block commutation approach to be easy and cost-effective from the point of view of hardware.

While the 120° block commutation offers simplicity, it is inherently associated with commutation torque ripple due to the abrupt switching transients and non-ideal trapezoidal BEMF. Vijayapriya et al. (2021) [6] addressed this performance limitation through a comparative study of different degrees of commutation. Their work investigated the trade-offs between the standard 120^{0} control and alternative methods, such as 180^{0} conduction, with the objective of quantifying the impact of commutation duration on torque ripple and overall motor efficiency. This research highlights the ongoing effort to optimize the balance between control simplicity and dynamic performance.

B. Regenerative Braking Systems for Electric Vehicles

The application of BLDC motors in EVs requires the creation of suitable regenerative braking systems which can offer maximum energy recovery and extend driving range. In this process, which includes the working of a motor like a generator to return energy to the battery, complex control logic needs to handle the energy flow reversal.

Singh, in turn, in 2022, discussed the problems of using the BLDC motor for implementing regenerative braking. The analysis showed that this method will require a sophisticated regulation mechanism to manage the stream inversion of energy. It is possible to perform this by switching

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stators in different configurations so that the motor operates in the generating quadrants of the plane. The basic theory underlines the control requirements will be complex to suit the needs of both safety and efficiency.

As an extension of the requirement of sophisticated regulation, Athira and Shenil (2021)[7] conducted research on new regenerative braking controllers. They proposed and compared two advanced techniques for this purpose: a Sliding Mode Controller and an Adaptive PI Controller. Their results, obtained through simulations in MATLAB, proved that the nonlinear SMC gave superior results compared to the Adaptive PI technique, especially regarding better control while decelerating and ensuring effective energy return. This indicates that in modern electric vehicle applications, which involve system nonlinearities and dynamic changes while the braking is in process, significant benefits arise from controllers capable of handling such phenomena.

While extensive research confirms the efficacy and implementation feasibility of fundamental trapezoidal control for BLDC motors, and separate studies have validated advanced non-linear control techniques (like SMC) for regenerative braking, a persistent gap remains in integrating these optimal control strategies with the detailed dynamics of an entire vehicle system, such as the quarter-car model demonstrated in the preliminary project files. This work seeks to **synthesize the proven simplicity of trapezoidal commutation with an optimized regenerative braking strategy**, while rigorously modelling the entire drive system within the Simulink environment to provide a cohesive solution for a specific EV application.

III. BLDC MOTOR

A. Torque in BLDC motor

The generation of continuous, steady electromagnetic torque in a BLDC motor[2] relies on the precise temporal and spatial alignment of the applied phase currents with the trapezoidal BEMF.

The fundamental expression for electromagnetic torque is related to the instantaneous power conversion:

$$T_e \propto \sum E_{BEMF} . i$$

E_{BEMF} is the back EMF of phase and I is the phase current.

To maximize and stabilize the torque output, the controller must ensure that the applied phase current waveform is a block wave (quasi-square wave) that is spatially and temporally synchronized with the flat-top region of the trapezoidal BEMF. By driving only two of the three phases at any given time (one high, one low, and one floating), and ensuring the block current pulse aligns with the constant BEMF region, the product EMF and Flux is close to a constant maximum value, minimizing torque ripple.

This alignment strategy is the foundation of the simplified 120-degree commutation process, which uses position sensors (typically Hall-effect sensors) to track the rotor position and sequence the phase switching accordingly.

B. Mathematical Modeling

The BLDC motor model is simplified by assuming ideal trapezoidal BEMF waveforms and identical phase windings (Wye-connected). The fundamental voltage equation for any phase is[3][4]:

$$V_x = Ri_x + L\frac{di}{dt} + e_x$$

Where:

- V_x , i_x , e_x are the phase voltage, current, and BEMF, respectively.
- R is stator resistance.
- L is the self – inductance of the phase winding (mutual inductance is typically neglected in this model).

The BEMF e_x is proportional to the rotor speed and a function of the electrical angle (θ_e) :

$$e_x = k_e \omega_r f_x(\theta_e)$$

Here, k_e is the back EMF constant
ω_r is the motor's mechanical speed.

The instantaneous electromagnetic torque T_e is given by the sum of the power products of BEMF and phase currents divided by mechanical speed ω_m :

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m}$$

The mechanical dynamics of the drive system are described by:

$$T_e - T_L = J\frac{J\, d\omega_m}{dt} + B\omega_m$$

Where T_L is the load torque, J is the moment of inertia, and B is the coefficient of viscous friction.

IV. TRAPEZOIDAL CONTROL

Trapezoidal control is a simple, digital way of driving a BLDC motor. It does not involve the complex coordinate transformations (Clarke/Park) required by FOC.

A. Principle of Operation

The control strategy divides the electrical cycle into six distinct sectors. In any given sector ,only two out of the three motor phases are energized while the third phase is left floating, meaning not connected [3].

This is why it is often called 120-degree conduction.

- **Conduction Mode:** The inverter switches are controlled such that two phases are conducting current and the resulting current path supplies the required magnetic flux to pull the rotor along.
- **Commutation:** Commutation, or switching, of the active phases takes place every. This is usually triggered by Hall effect sensors embedded in the motor, which detect the rotor's magnetic pole.

B. Implementation of Control Loop

- **Hall Sensor Block:** Feeds back rotor position information to the controller.
- **Commutation Logic Block:** A state machine or look-up table that converts the 6-state Hall signal into the 6 corresponding Pulse Width Modulation PWM signals for the inverter switches.
- **Speed Controller:** This contains a closed-loop control system that involves, usually, a PI controller in outer loop that controls the speed of the motor by setting the duty cycle of the active PWM signal.

C. Speed and Control Loop Design

In most cases, outer speed control is performed using a Proportional-Integral (PI) controller:
Generate the necessary torque demand-which is proportional to the required current magnitude. The PI controller transfer function :

$$G_{PI}(s) = K_P + \frac{K_i}{s}$$

Proportional Term (K_P): The output is proportional to the instantaneous speed error. It improves the response time and reduces steady-state error.

Integral Term (K_i/s): Eliminates the steady-state error by accumulating the error over time. The magnitude of the reference current is the output of the speed controller, which is directly proportional to the duty cycle applied to the high-side switch of the conducting phases. Tuning the and the value of is usually chosen so that a fast response time with minimum overshoot, often using techniques like Ziegler-Nichols or pole placement.

V. REGERATIVE BRAKING IMPLEMENTATION

Regenerative braking in BLDC motor-driven electric vehicles enables the conversion of kinetic energy into electrical energy during braking and later stores back to the battery. Implementation of this strategy requires careful logic design to switch from motoring to braking mode based on pedal input and vehicle speed.

A. Control Logic

The core of regenerative braking is the power electronics dynamic reconfiguration along with the inverter switch gate signals adjustment. The control subsystem, which is illustrated in the simulink model, constantly checks the brake pedal (or command input) and current flow to ensure that regenerative braking is activated only during safe operational conditions.

The Brake Control block takes the braking command as input and through the use of logical gates and decision blocks that operate like relays, it sets the system to either motoring or braking state.

Hall sensors and back-EMF signals are decoded in dedicated subsystems to provide accurate rotor position for correct phase commutation during energy recovery.

Simulink uses conditional routing to safeguard the switching between motoring and braking modes by monitoring the direction of current and speed thresholds.

B. Simulink Model

To implement the model, blocks are used for logic, gating, and subsystem integration:

- In the Regenerative Braking Control part, the required current setpoint is calculated during braking and gate pulses are modulated to allow controlled energy flow back into the battery.
- The Switching Logic guarantees that the inverter arms are set appropriately for braking according to the real-time EMF and Hall sensor conditions.
- Moreover, extra comparators and relay blocks provide protection against hardware malfunctions or hazardous operating points, as illustrated in the Simulink diagrams.

C. Operating Principle

During braking process, the BLDC motor acts as a generator, and the induced back-EMF creates a current that is fed into the battery. The relation is governed by:

$$P_{regen}=e_{phase}\cdot I_{regen}$$

where P_{regen} is the recovered power, e_{phase} is the phase back-EMF, and I_{regen} is the braking current.

The Simulink gating ensures that this current is permitted only in the designated braking sequences, restricting energy flow for maximum safety and efficiency.

D. Control Challenges and Debugging

It is a big task to develop the motoring and regenerative braking modes transition to be seamless:

- PWM interaction: If the timing is not matched correctly, it may lead to low current or chattering at slow speeds.
- Sensor logic: If the Hall or EMF information is decoded incorrectly, it can result in wrong switching.

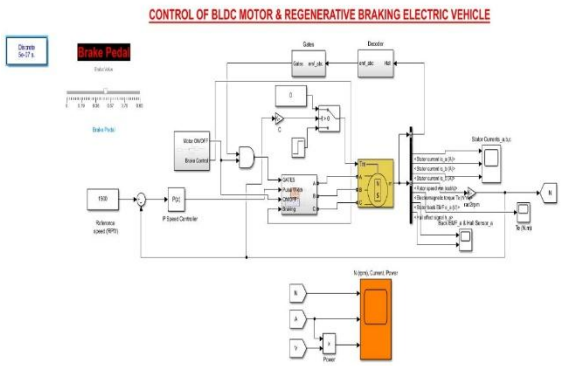


Fig.1 Simulink Model

- Tuning issues: The control parameters like PID gains need proper adjustment for stable current and torque during braking.

Currently, simulation results are still pending as the team is debugging subsystem interactions and optimizing controller parameters. Typical problems are unstable torque produced in the braking mode and inverter behaving unexpectedly when switching states.

VI. SYSTEM DESIGN AND INTEGRATION

A. Simulink Architecture Overview

Simulink is the tool of choice for designing the entire system, which is made up of the discrete blocks for the speed reference, PI control, braking logic, commutation, and switching of power electronics. Everything that has to do with parameters, signals, and subsystems is virtual, corresponding to a digitally simulated BLDC drive and regenerative brake scenario.

B. Control Subsystem Integration

Simulation signals provide the interconnection between the subsystems. The speed control and commutation follow logical blocks, while the braking function is activated by the simulated brake command. The mode-switching is realized through the digital signal conditions in the Simulink environment with no actual input from actuator or sensor.

C. Transition Logic

The switch from driving to regenerative braking is implemented through logical gate and relay blocks in the Simulink modeling environment. All scenarios and safeguards depend on virtual measurements and logic states.

VII. SAFETY, FAULT HANDLING, AND LIMITATIONS

A. Embedded Protection Routines

Protection and diagnostics are handled by means of thresholds, comparators, and logic blocks embedded in Simulink. Inside the simulation, over-current and under-voltage events, as well as shoot-through prevention, are treated and others diagnostics are made leading to the change of model state in a safe way.

B. Debugging Challenges

The work of debugging is concentrated in the area of signal timing, logical conflicts, control loop tuning, and stability problems within the simulation, rather than in physical troubleshooting.

C. Simulation Limitations

The limitations are in model simplifications (ideal measurements, ignoring of noise), and in the restrictions of the representation of real-world anomalies. Performance artifacts and bugs are specific to the simulation.

VIII. FUTURE WORK

The outcomes of the simulation and the ongoing progress of the BLDC motor regenerative braking model leads to different applications being investigated and refined for the future.

- Refinement of Model: Adding a few more complexities to a model is possible if simulation of the finer points has been allowed in simulating features like electrical circuit noise, sensor inaccuracies, and non-ideal switching behaviours. Those improvements will give an opportunity for more vigorous tests in the presence of different kinds of disturbances to the system.
- Enhancement of Control Algorithm: Installing an adaptive or fuzzy logic controller for both the speed and brake could stabilize the entire system even more during the transient events. Machine learning approaches might make it possible to optimize control parameters with less manual trial-and-error through data-driven techniques.
- Fault Scenarios that are Wider: Future studies are necessary to assess the more diverse fault conditions in the simulated environment such as sensor dropout, signal timing jitter, and simulated electrical shorts. This would increase the fault tolerance and diagnostic capabilities of the control system.
- Analysis of Parameter Sensitivity: A thorough sensitivity study of the model to parameters like stator resistance, inductance, and back-EMF characteristics would result in a better understanding of the trade-offs among performance and reliability.
- Models for conserving energy through the integration of electric vehicles: the Braking simulation coupled with the battery state-of-charge and thermal management sub-models could give a more complete picture of the energy flow at the vehicle level which would, in turn, support the further optimization of EV braking and drive efficiency.

To sum up, there is a very significant scope for more work to be done on the simulation-only approach before the hardware

prototyping is even considered, and each one of the suggested areas would serve more secure and efficient electrification in

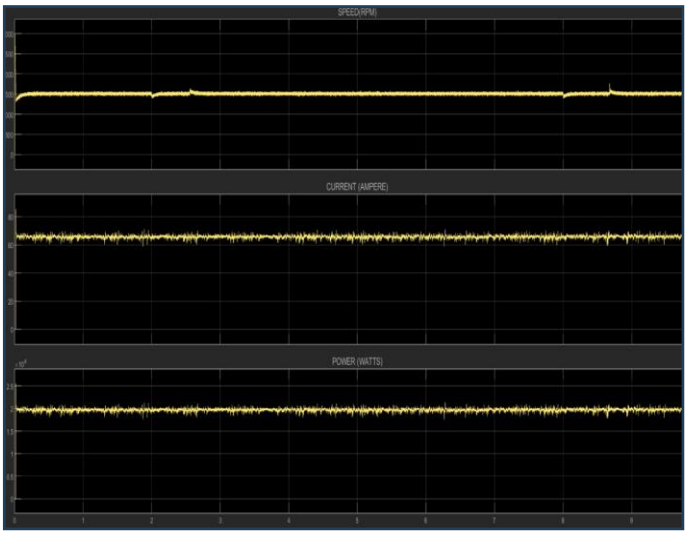


FIG 2. GRAPHICAL RESULT OF SIMULATION- SPEED, CURRENT AND POWER

IX. CONCLUSION

A. Summary of Work This paper shows a complete digital model and control technique for a BLDC motor drive and regenerative braking, relying entirely on a simulation-based strategy in the Simulink environment. The design process consists of speed reference generation, dynamic PI speed control, motor commutation logic, and advanced energy recovery methods through regenerative braking.

B. Technical Achievements The model, through modular subsystem development, accurately depicts the switching from motoring to braking and vice versa through the use of logic control, sensor signal decoding, and timed switching algorithms. During longer simulation runs, the simulations of protective features, fault handling routines, and safety protocols were implemented as blocks.

C. Simulation-Based Insights Although the project is still in the debugging phase and does not have any measured results, a few simulation-based insights have been obtained: *The flow of logic needs to be very controlled so that there is no overlap between modes and no unintentional switching events.

*The accuracy of sensor signals (like Hall and back-EMF emulation) is of utmost importance for the correct commutation and recovery of energy.
*Protection mechanisms are essential even in simulated environments to prevent unrealistic results or system lock-up during test scenarios.

D. Broader Impact and Applicability

The modeling and control approach demonstrated through this study provides a solid base for improvising on the virtual prototyping of EV subsystems, exploration of advanced control techniques, and even scaling up to more complex drive and braking systems. The simulation's flexibility and capacity to be extended give room for the researchers to change the model for new algorithms or just to use it for more extensive testing conditions without any hardware limitations.

E. losing Remarks

This research has made a significant contribution to the field of BLDC motor regenerative braking control through the combination of subsystem logic, simulation-based protection, and digital algorithms in one extensive Simulink model. Furthermore, it has demonstrated the potential of simulation for speeding up the process of developing and validating energy-efficient vehicle drive strategies. It is anticipated that the testing of future results after debugging will not only reinforce the validity of the approach but also realize its benefits for EV controller design.

X. ACKNOWLEDGMENT

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XI. REFERENCES

[1] Fundamental of Electric Drives by GK Dube
[2] Electrical Machines by BL Theraja
[3] Power Electronics and Drives by Bimal K Bose
[4] Electrical Machinery by PS Bimbhra
[5] C. R. E. S. Rex, Y. Gayathri, K. Neelima, and K. Bhargava, "Speed Control of BLDC Motor Using Trapezoidal Commutation & PWM Technique," in *2023 4th IEEE Global Conf. for Advancement in Technology (GCAT)*, Bangalore, India, 2023.
[6] P. Vijayapriya, A. Thamilmaran, A. Bhatnagar, and N. A. Reddy, "Comparative Analysis on Control Techniques of a PMBLDC Motor at Different Degree of Commutation," in *2021 Innovations in Power and Advanced Computing Technologies (i-PACT)*, 2021. doi: 10.1109/I-PACT52855.2021.9696855.
[7] M. Athira and P. S. Shenil, "Novel Regenerative Braking Controllers for Electric Vehicle Driven by BLDC Motor," in *2021 4th Int. Conf. on Electrical, Computer and Communication Technologies (ICECCT)*, 2021. doi: 10.1109/ICECCT52121.2021.9616660.
[8]U. K. Singh, "Analysis of A Regenerative Braking System of BLDC Motor In Targeting Electric Vehicle Implementation," in *2022 4th Int. Conf. on Emerging Res. in Electronics, Computer Science and Technology (ICERECT)*, 2022. doi: 10.1109/ICERECT56837.2022.10060185.

[9] C. Jhawar, U. B. Manthati, and P. Abhishek, "Design and Implementation of Sensor-based Trapezoidal Control of BLDC Motor based on XMC-7200 Microcontroller for EV Application," in *2024 Int. Conf. on Computer, Electronics, Electrical Engineering & their Applications (IC2E3)*, 2024. doi: 10.1109/IC2E362166.2024.10827162.

XII. BIOGRAPHY

Dr. Subramaniam K works in the department of Electrical Engineering at Vellore Institute of Technology. Their responsibilities include teaching and supervising research projects in the area of Power Electronic and Drives. Dr. Subramaniam K assisted with guidance during the preparation of this research paper.

Abinaya S is a third year B.tech Electrical and Electronics Engineering Student at Vellore Institute of Technology. she contributed significantly to the development and simulation of the trapezoidal control algorithm for the BLDC motor and the comparative analysis of the regenerative braking controllers. Ms. Abinaya plans to pursue graduate studies in Electric Vehicle Technology and aims for a career in R&D within the automotive or power industries.

Hannah Ajish is a third year B.tech Electrical and Electronics Engineering Student at Vellore Institute of Technology. she contributed significantly to the development and simulation of the trapezoidal control algorithm for the BLDC motor and the comparative analysis of the regenerative braking controllers.